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Targeted Protein Degraders (TPD)
Case Study-2

TPDs at APSL

- Very experienced synthetic chemistry team with Drug Discovery experience.
- Dedicated analytical support (Prep. HPLCs, RP MPLCs, Buchi L-300 Lyophilizer etc.).
- Analytical team with significant experience in the purification of TPDs.
- Rapid and flexible access to commonly used reagents, >40000 in-house chemicals.
- Well-structured and confidential documentation processes, inclusive of patenting.
- Smooth shipment processes with a proven global delivery track of <5 days.

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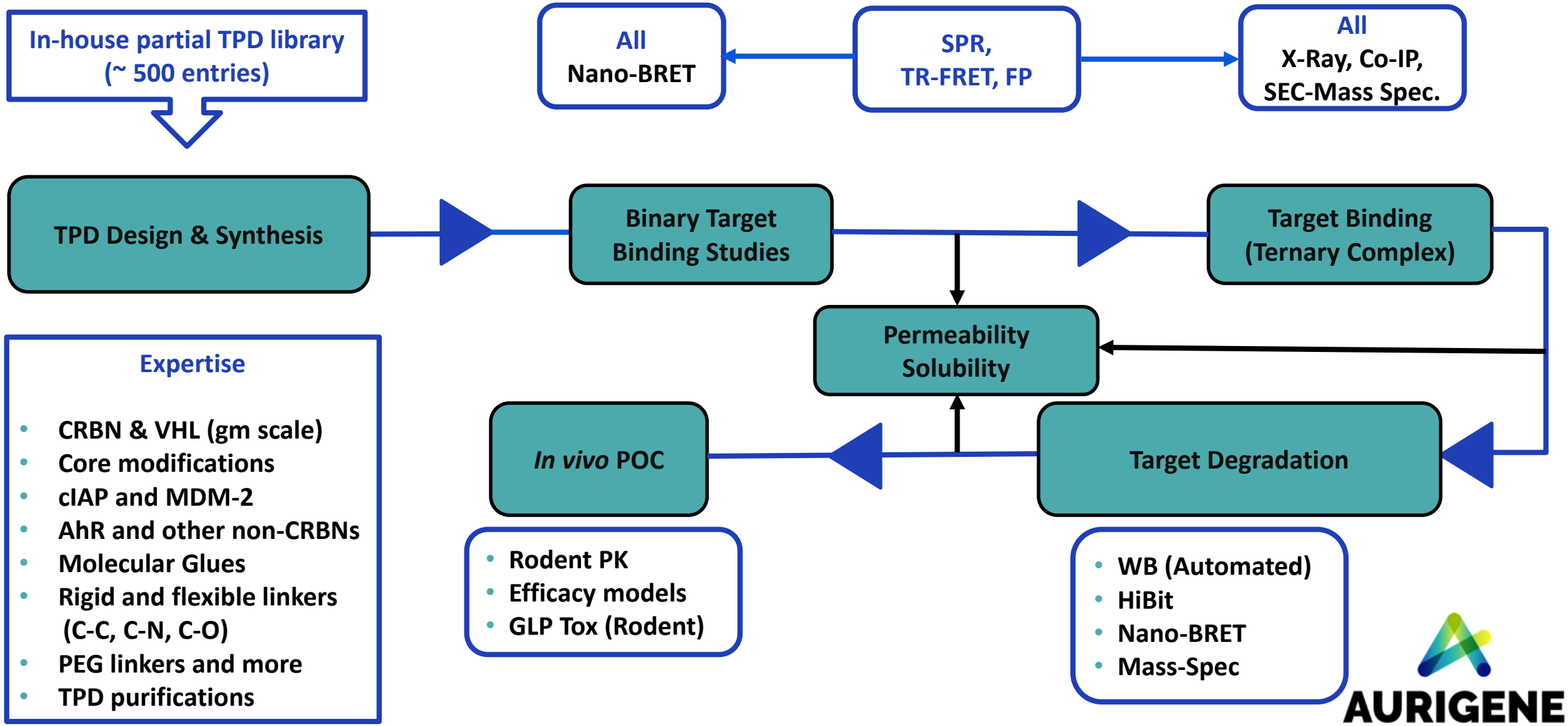
TPD – Workflow

TPD Synthesis and Various Screening Modules at APSL



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TPD Capabilities at APSL

Synthetic Chemistry

- Various C-C and PEG-Based Linkers (n = 4-12 and beyond)
- Various-NH_2, -CO_2H, -N_3, -CCH and –X and –OH terminal Linkers
- Diversely substituted CRBN Ligands (gm scale)
- Various VHL-Ligands for linkage through –NH_2 or -OH
- Tailor-made partial TPDs ready for target linkage
- Synthesis of standard cIAP and MDM-2 Ligands
- Synthesis of non-CRBN/VHL E3-Ligase ligands

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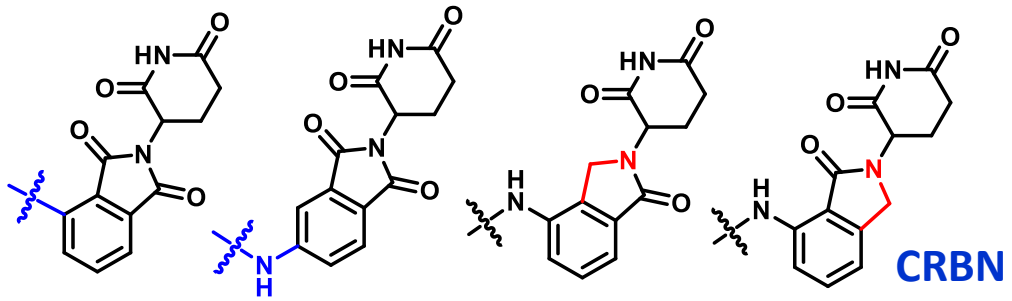
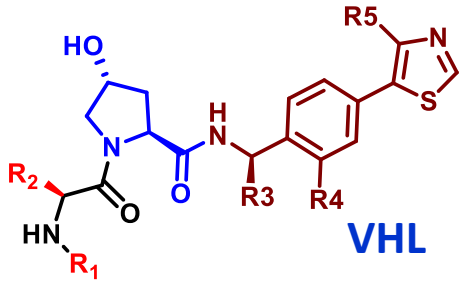
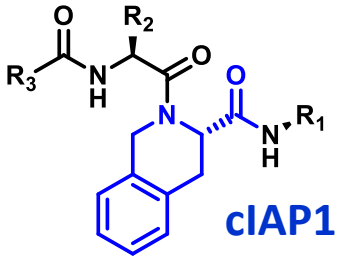
Synthesis of E3-Ligase Ligands & Linkers

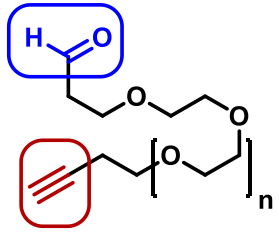
- Synthesis of various E3 ligase ligands like CRBN inhibitors (Thalidomide, Pomalidomide and Lenalidomide analogues, C4, C5, C6 or C-7 versions)
- Various E3 ligase ligands that are inhibitors of MDM2 and cIAP1
- 4-Hydroxyproline analogues that are selective inhibitors of VHL

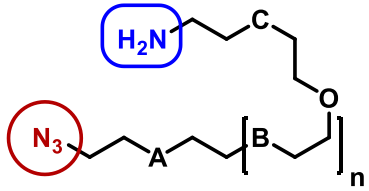
- Synthesis of various PEG- or alkane-related bivalent cross-linkers
- Tunable end-groups including –CO2H, -NH2, -N3, -alkyne, -CHO, -halides and more
- Expertise in ligand-PEG attachments using C-C, C-N, C-O and C-P linkage strategies
- Extensive drug discovery experience in designing the partial TPDs “as needed”



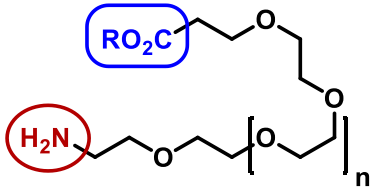


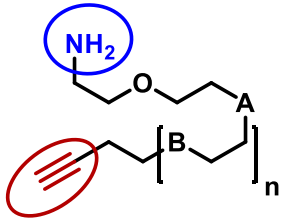




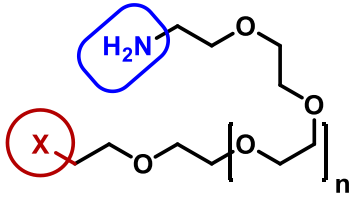


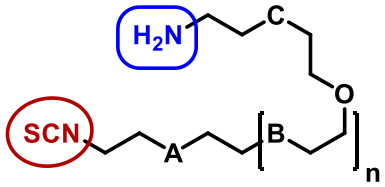
A, B, C = CH_2, -NH, -N-Alk or -O





A, B, C = CH_2, -NH, -N-Alk or -O





A, B, C = CH_2, -NH, -N-Alk or -O

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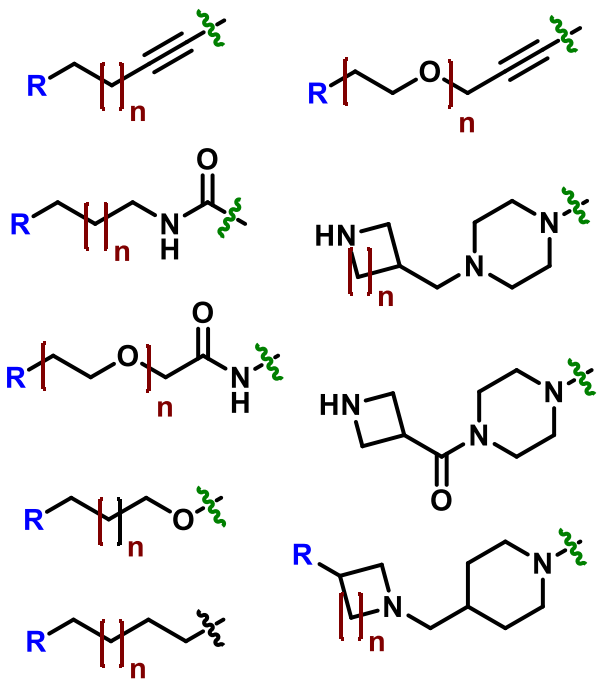
Possible TPDs for POI Using In-House Library

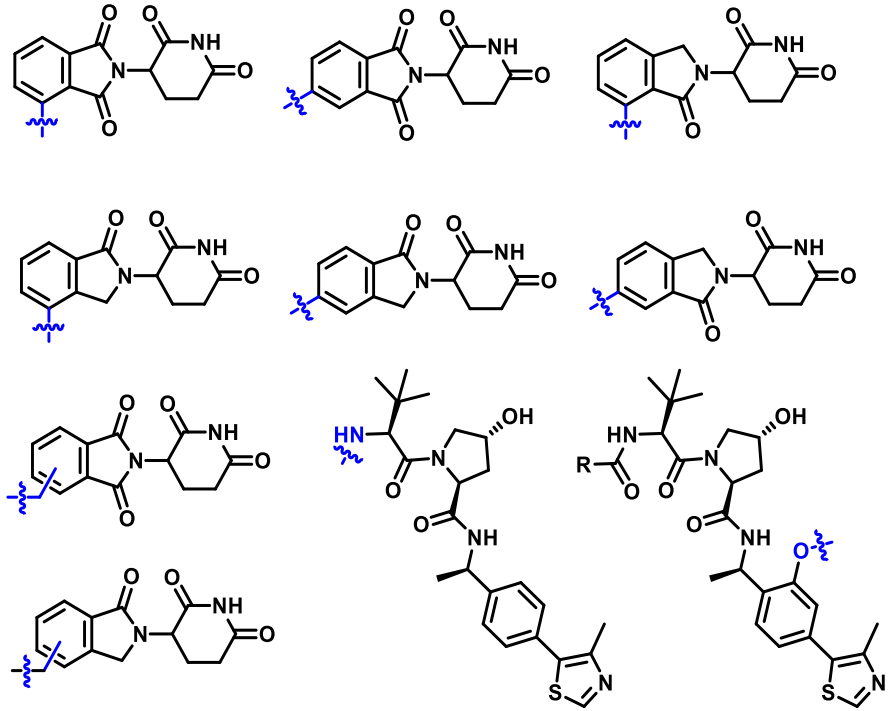
POI-Ligand — Linker of choice — E3-Ligase Ligand

POI-L from Collaborator +



R = NH_2, CO_2H, NCO, N_3, OH, -X etc.

+



- Dedicated Analytical Support
- Dedicated prep. HPLC system, Lyophilizer etc.
- Optional CADD Support

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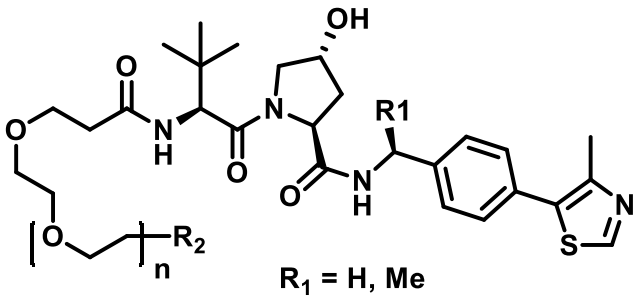
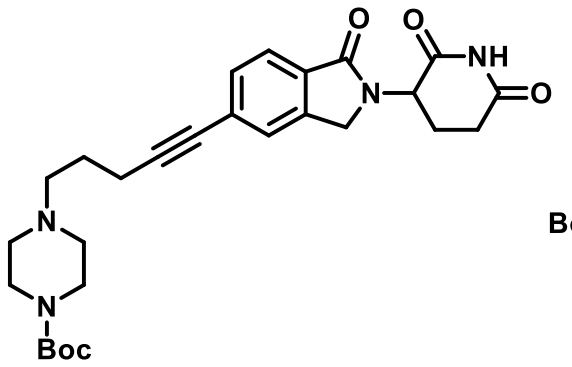
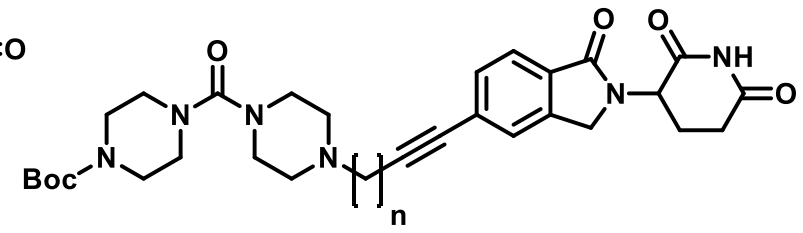
Partial TPDs: Representative Examples

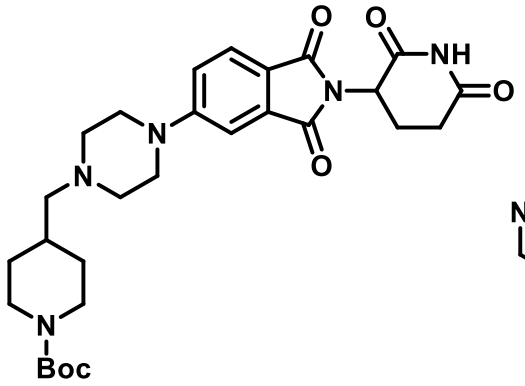


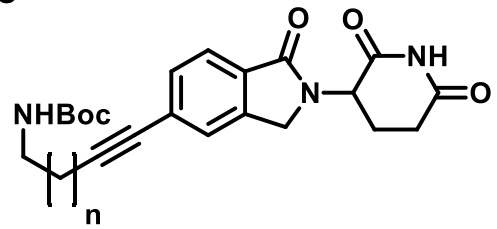
R_1 = H, Me

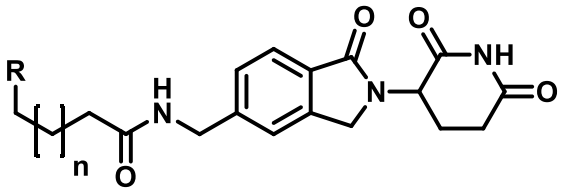
R_2 = NHBoc, CO_2tBu



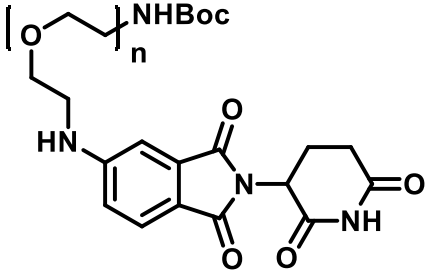








R = NHBoc, CO_2tBu



~ 500 Entries (E3-Ligands, Linkers and partial TPDs) and growing …

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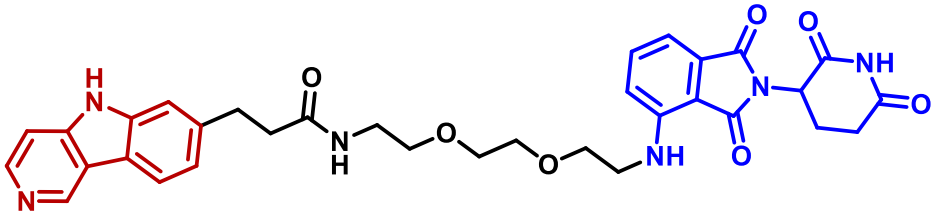
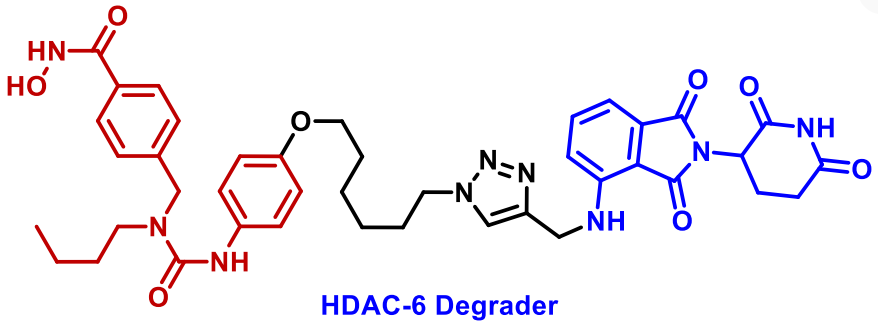
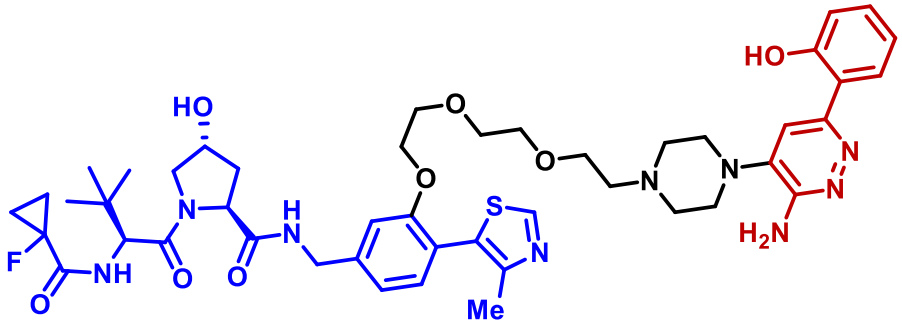
Synthesis of Various Published TPDs



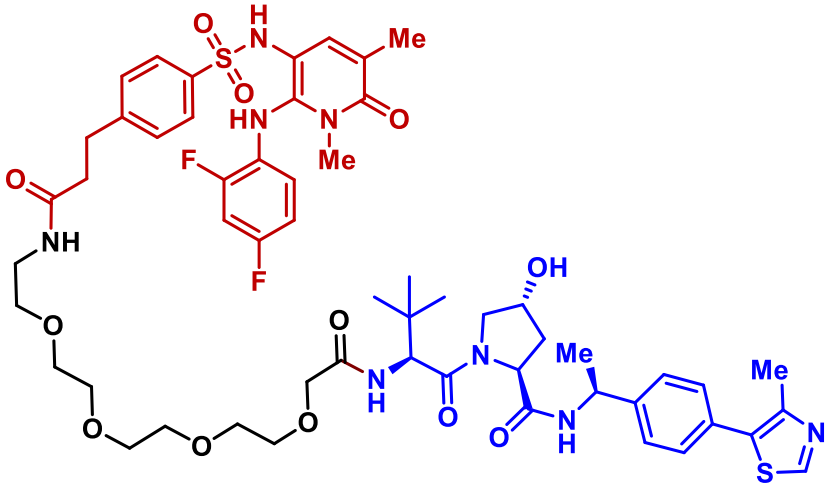
Aberranat-Tau-Degrader QC-01-175
Nathanael S Gray *et al*
eLife 2019, *8*, e45457



HDAC-6 Degrader
Weiping Tang *et al*
J. Med. Chem. 2019, *62*, 7042-7057



SMARCA2 Degrader BI-2660
Boringer Ingelheim
Nat. Chem. Biol. 2019, *15*, 672–680



MEK Degrader: 90 % Degradation @ 1 μM (16 h)
PROTAC 3: J. Med. Chem. 2020, 63, 157-162

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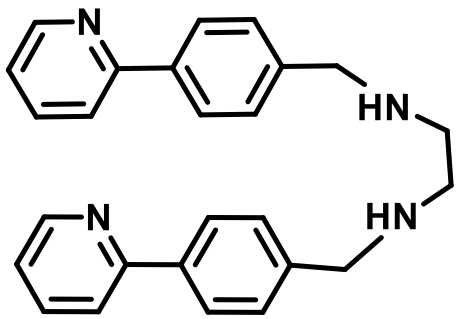
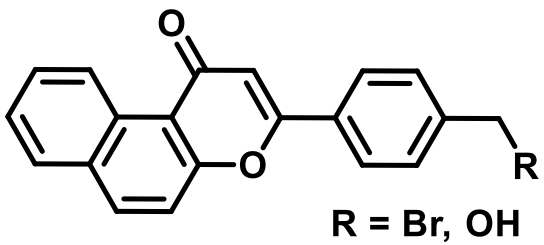
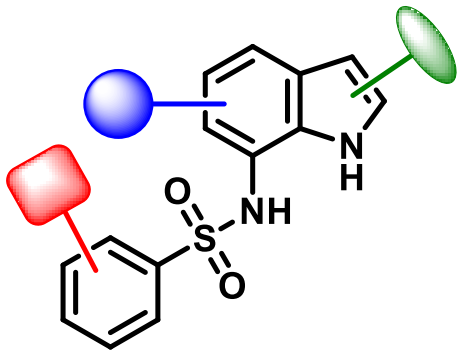
Non-Standard E3-Ligase Ligands



F-box protein 3 Inhibitors



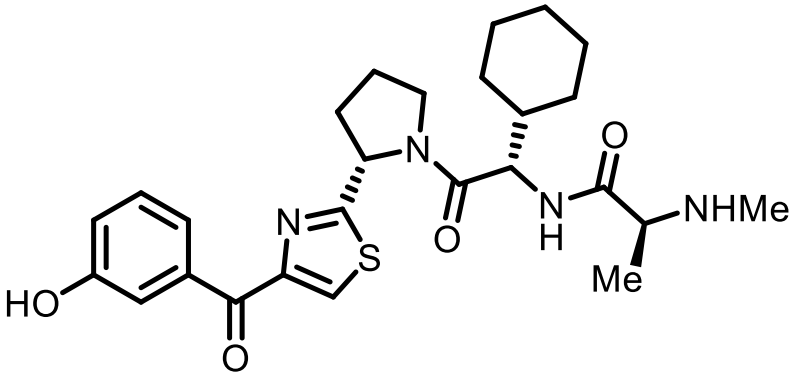
AhR Ligand (α-NF, β-NF)



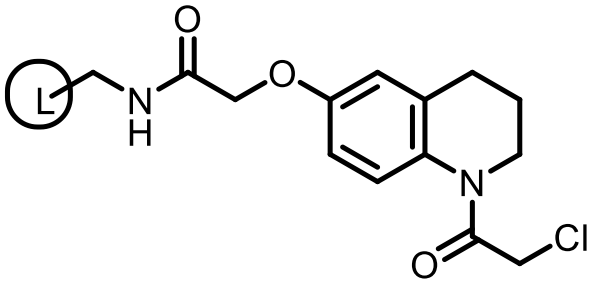
RBM39 Degraders

Nature Chem. Biol. 2020, 16, 15-23

Indisulam Analogues



cIAP Ligands (Boc)



KB-02-CO_2H (DCAF16)

DCAF-hijacking degraders

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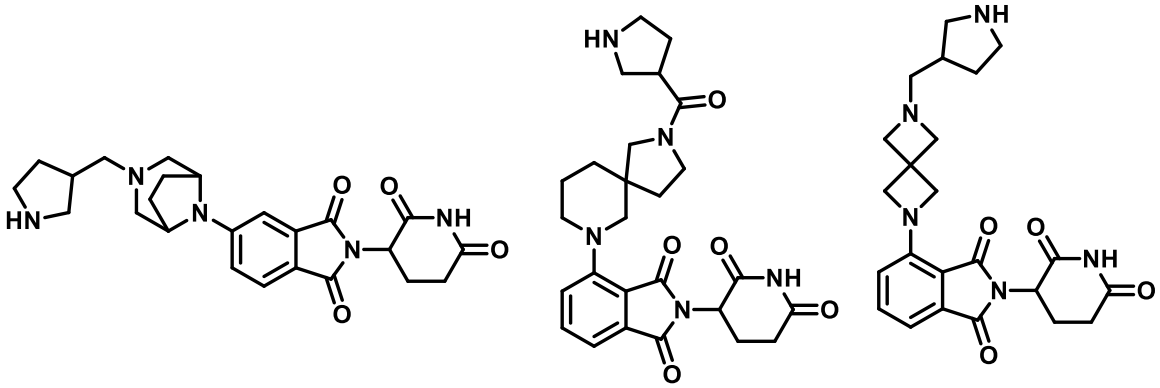
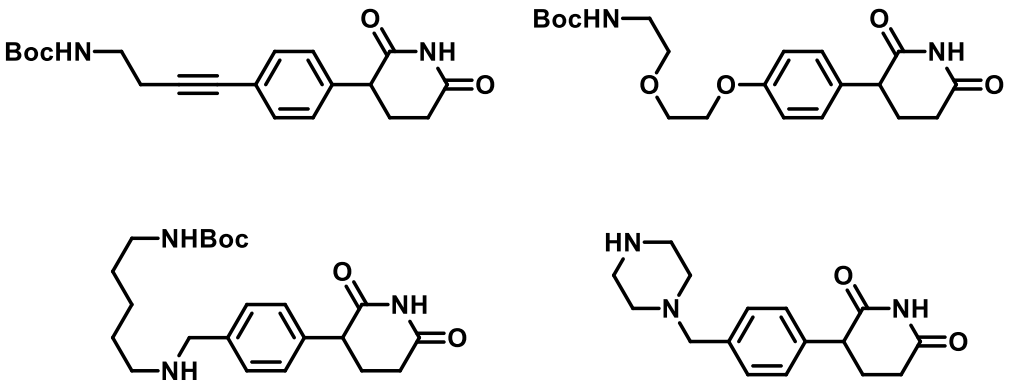
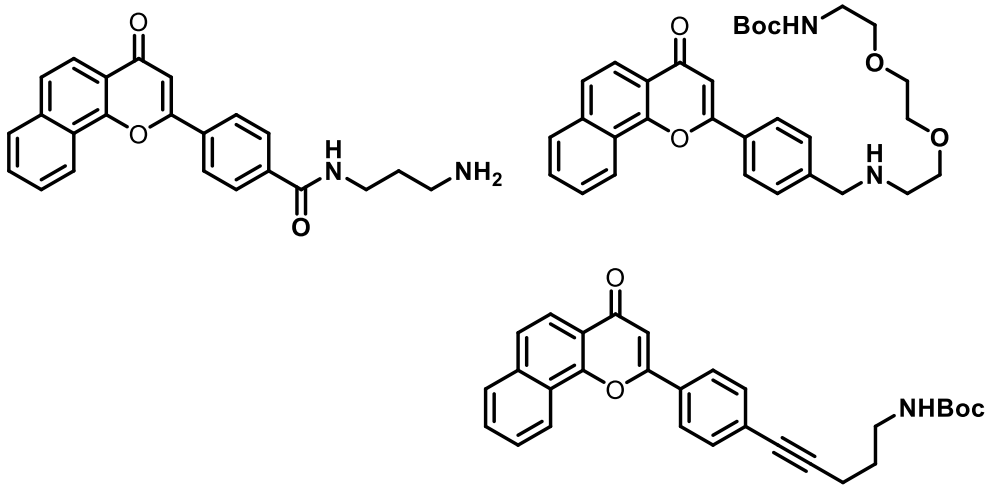
New Avenues and Endeavors



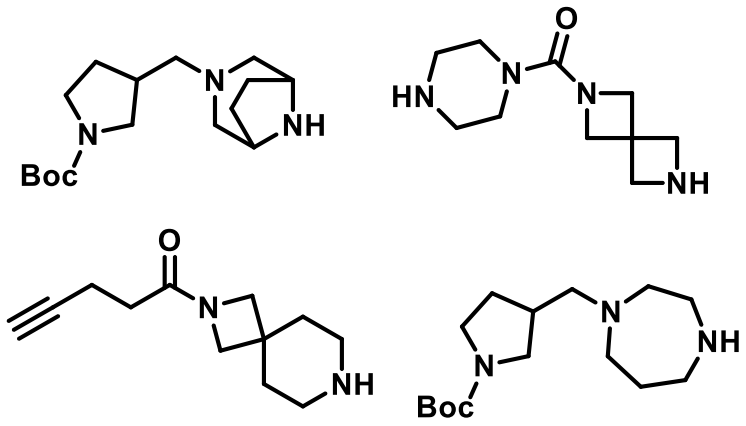
Novel Rigid Partial TPDs (~20 Entries and growing)



Modified CRBN-Based Partial TPDs (~10 Entries and growing)



AhR-Type Partial TPDs (Initiated)



Rigid Linkers for non-CRBN E3-LLs (~10 Entries and growing)

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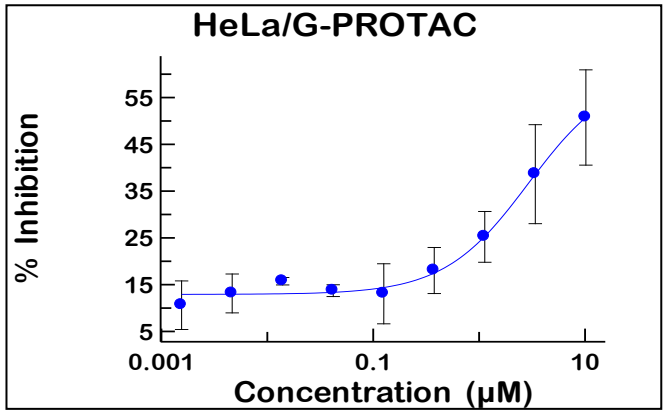
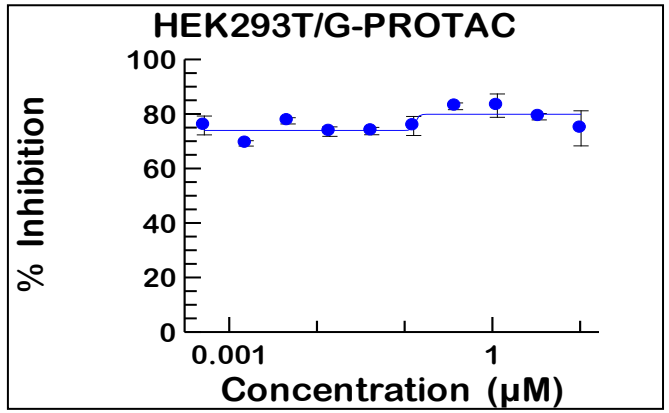
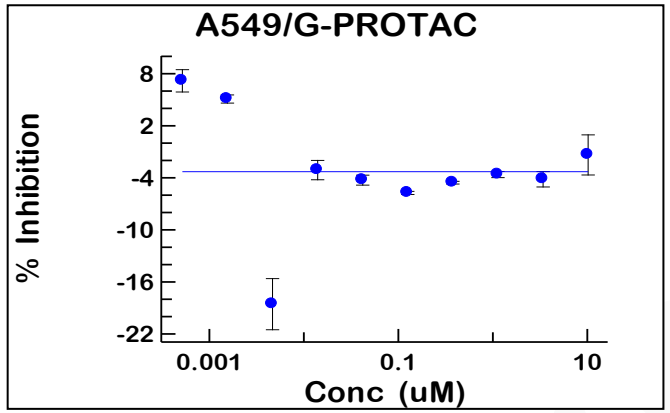
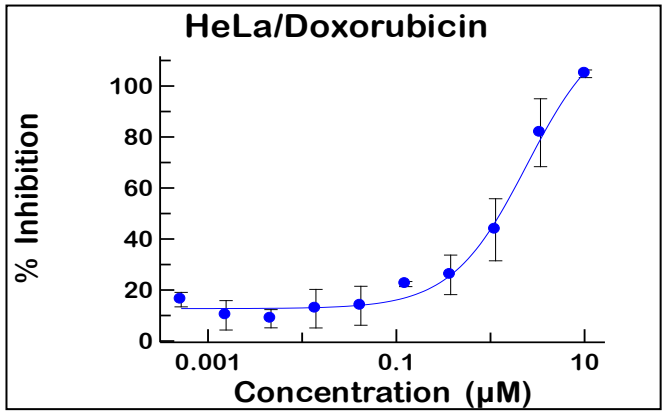
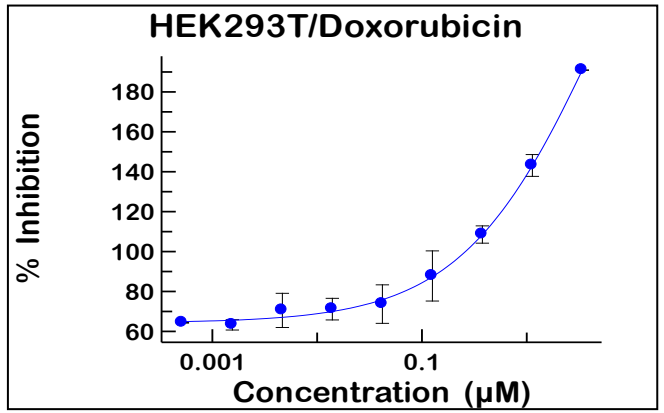
Gefitinib-Based TPD-3: Cell Proliferation Assays

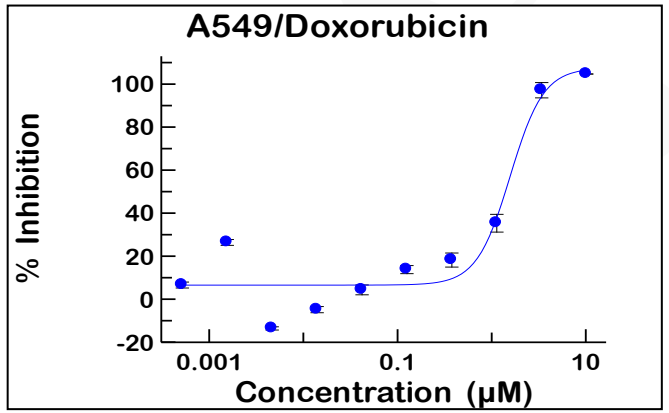


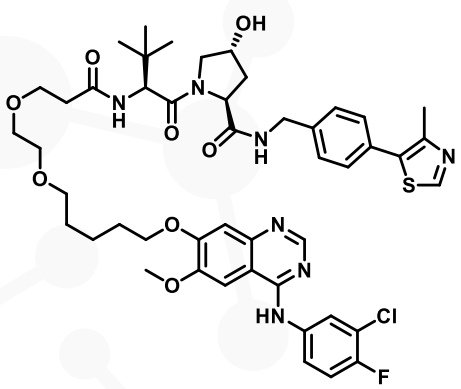












Gefitinib-Based TPD-3

G-PROTAC3 IC50 values (µM)	
HeLa	2.87
HEK293T	0.13
A549	NA (no inhibition)

Doxorubicin IC50 values (µM)	
HeLa	2.38
HEK293T	0.66
A549	1.54

Dose response with G-TPD3 observed only in HeLa cells

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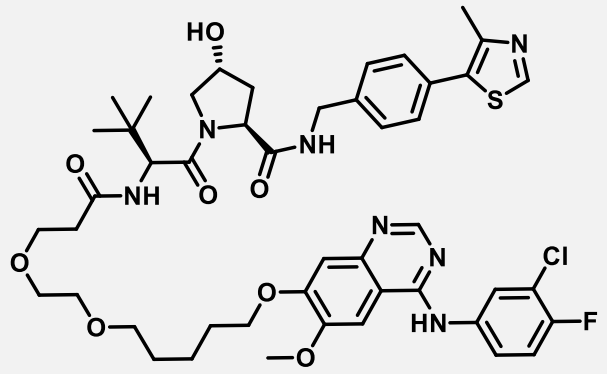
Profile of Gefitinib-Based TPD-3: ADME Data



Gefitinib-Based TPD-3

Study Details		
Prolif_A-549 IC_{50} (µM)		No
Prolif_HeLa IC_{50} (µM)		2.87
Prolif_HEK293T IC_{50} (µM)		0.13
Solubility (µM)	Thermodynamic pH 6.8	99
	Kinetic pH 6.8	17.6
Log D	Octanol:buffer pH 7.4	4.94
Plasma Stability: t ½ (min)	Rat	>240
	Human	>240

Study Details		
Hepatocyte Stability: CLint (µL/min/million cells)	Mouse	105.69 ($t_{1/2}$: 6.56 min)
	Rat	88.48 ($t_{1/2}$: 7.83 min)
	Human	60.42 ($t_{1/2}$: 11.47 min)
Metabolic Stability: t ½ (min)	Rat	>240
	Human	>240
Permeability: MDR1-MDCKII	A ►B Papp (10^{-6} cm s^{-1})	0.02
	B ►A Papp (10^{-6} cm s^{-1})	2.48
	Efflux Ratio	144
PPB (% bound)	Mouse	99.98 (<1.0 % Free)
	Rat	99.90 (<1.0 % Free)
	Human	99.80 (<1.0 % Free)
CYP Inhibition (µM)	Mouse	
	Rat	
	Human	

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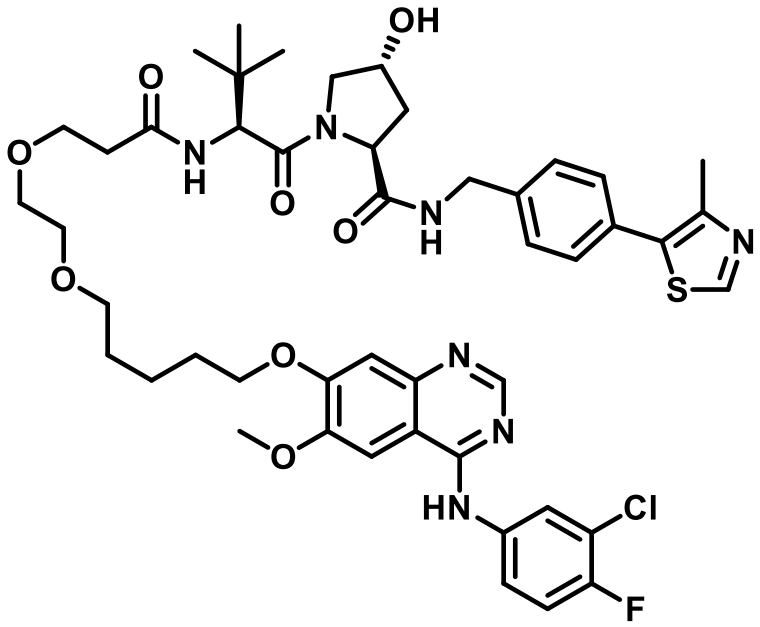
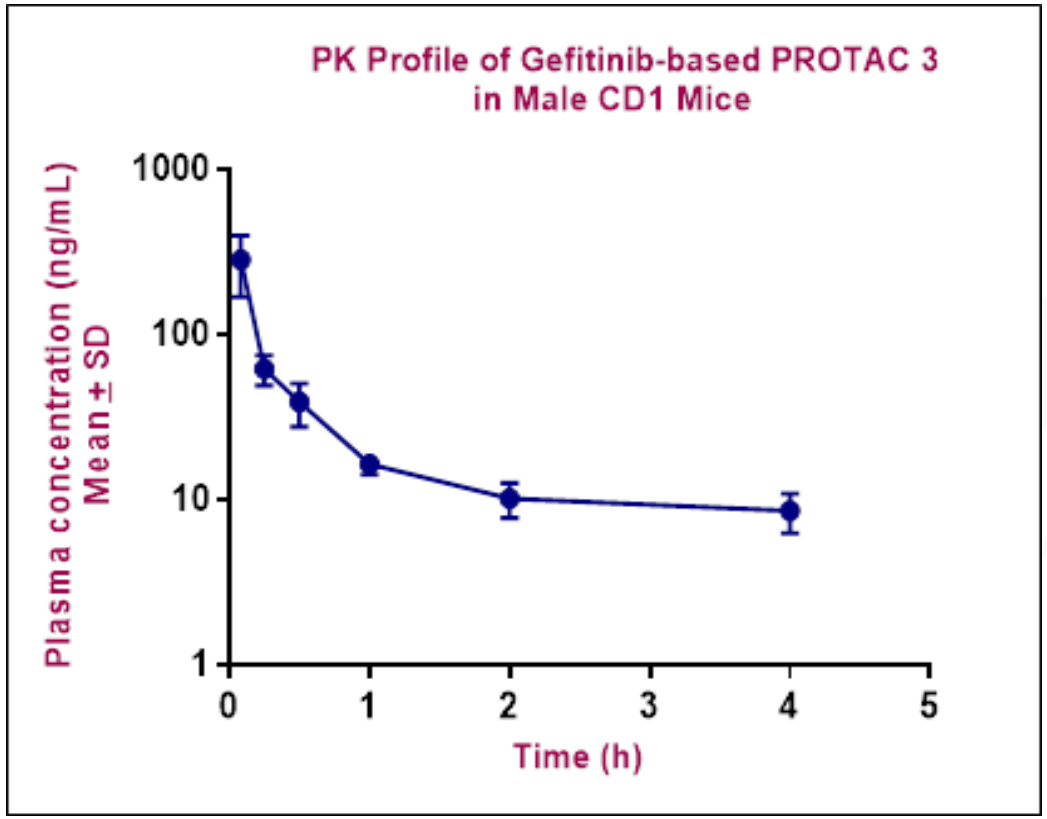
Gefitinib-Based TPD-3 – PK Profile



Gefitinib-Based TPD-3



Dose and Route:	3 mpk, Intravenous
Formulations:	NMP + PEG-200 (10:90)

i.v. PK profile in mouse

Mean PK Parameters									
Mean PK Parameters	Dose	C0	AUC_{last}	AUCINF_obs	$T_{½}$	Cl_obs	Vss_obs	Rsq	MRT
	mg/Kg	(ng/ml)	(ng*hr/mL)	(ng*hr/ml)	(hr)	mL/min/kg	L/kg		(hr)
	3	574.48	121.40	164.13	3.49	304.63	52.99	0.8093	0.74

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Case Study: Kinase1-CRBN-APSL-TPD-10003 Ternary complex

Kinase1-CRBN-APSL-TPD-10003 ternary complex model was predicted

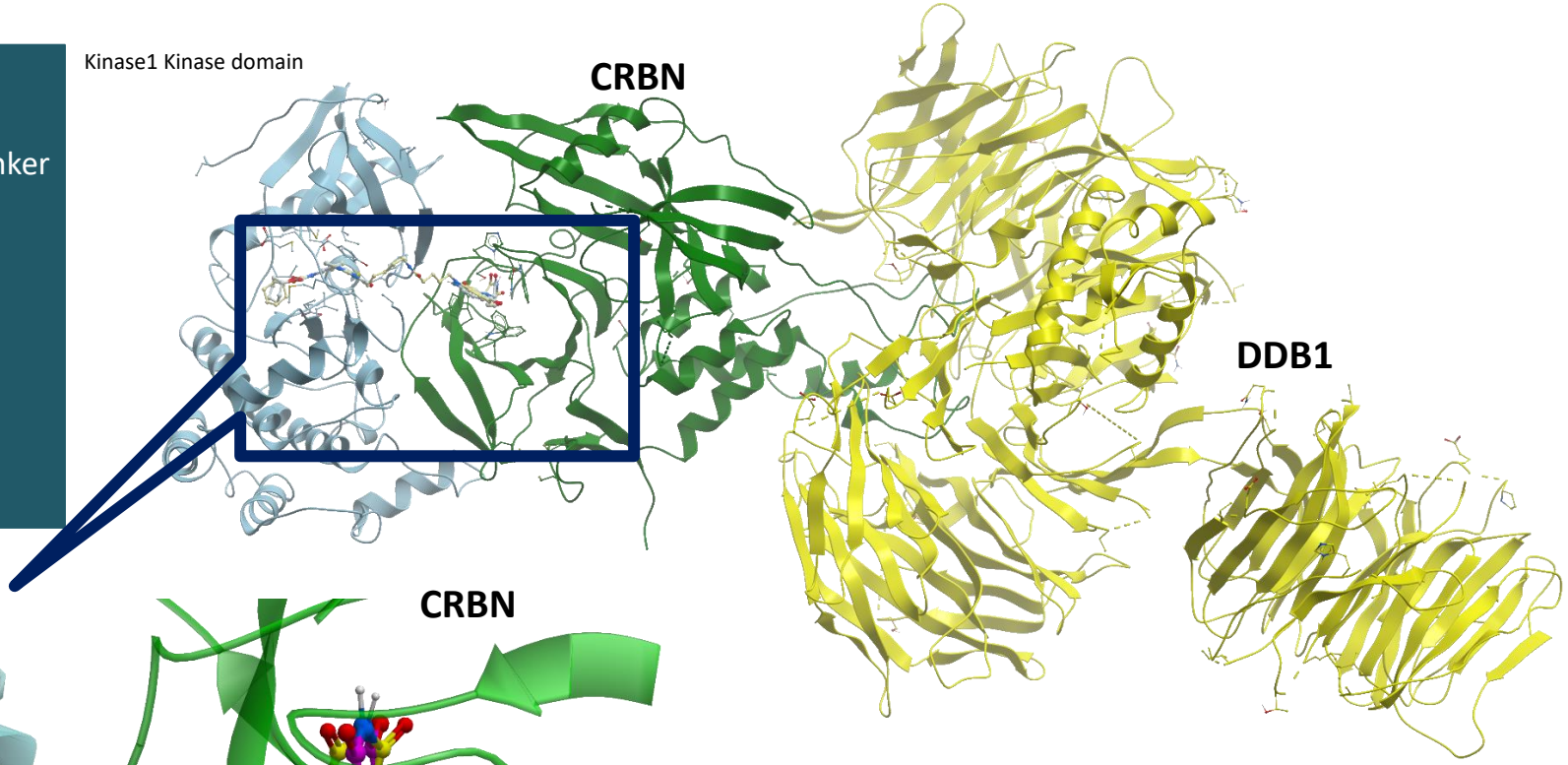
Solvent region extension from Kinase1 and CRBN with optimal linker length was identified.

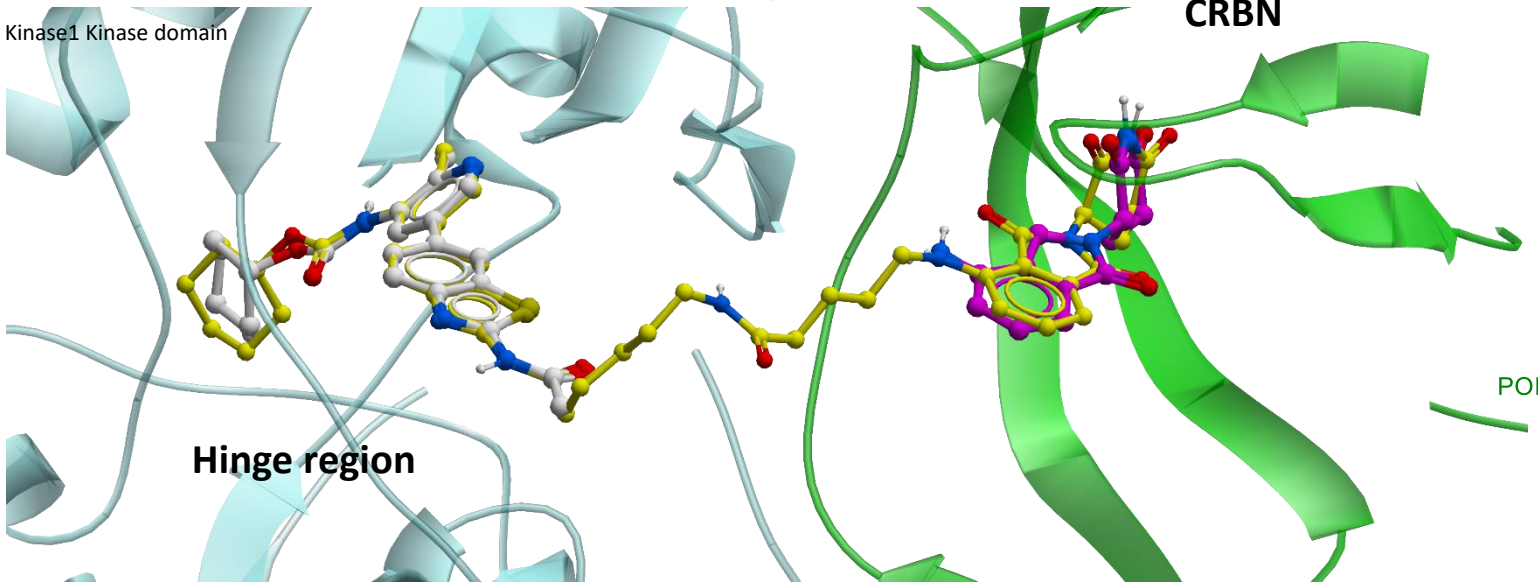
Orally bioavailable rigid linker may enhance overall property of -10003 based TPD.

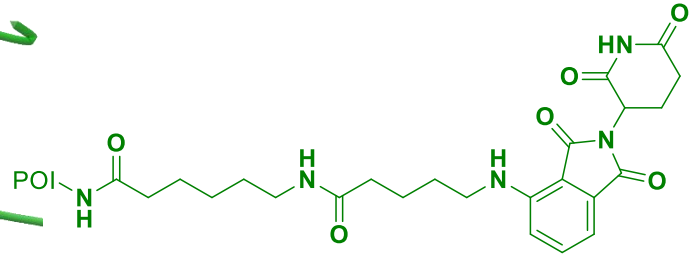
More rigid linker based on -10003 was designed

Total 15 TPDs were shortlisted for synthesis

10 TPDs were made and tested



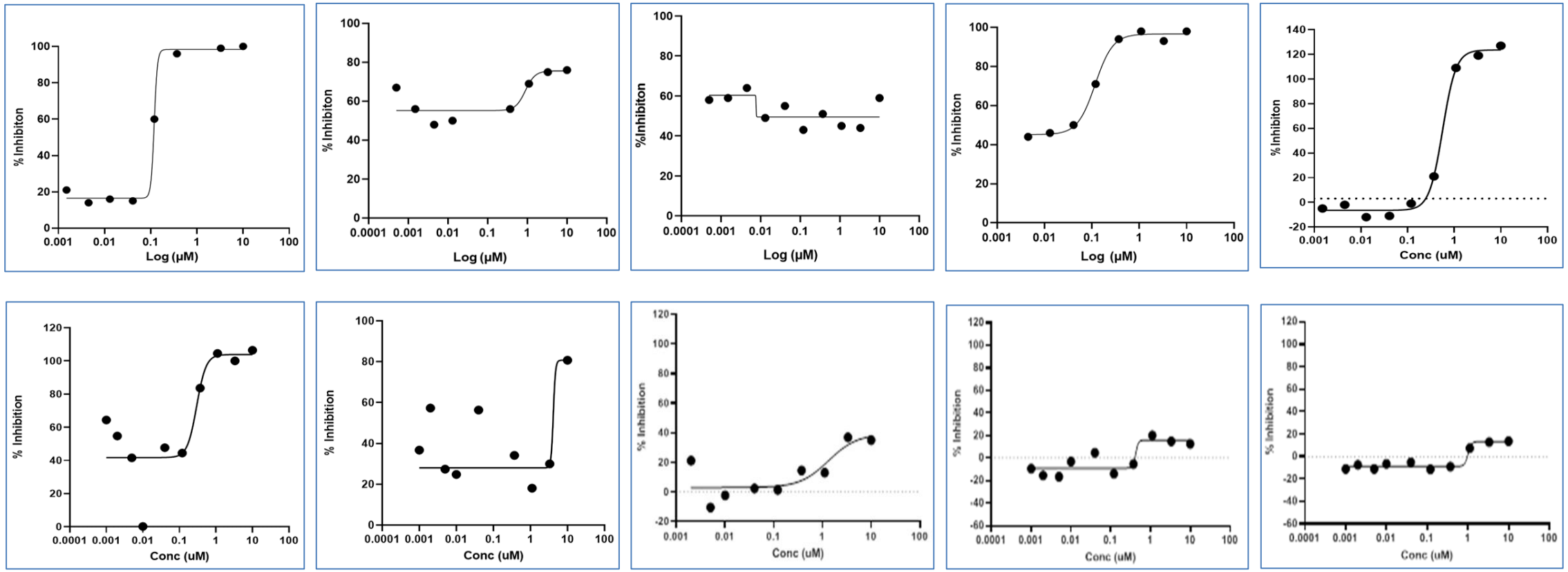




PROTAC-10003

APSL-TPD-10003

Case Study: *In vitro* Biology Capabilities – TPDs (Biochemical Assay)



- **10 TPDs were screened in biochemical assays**
- **TPDs showing >50 inhibition at 1 uM were tested in cell based functional assays**

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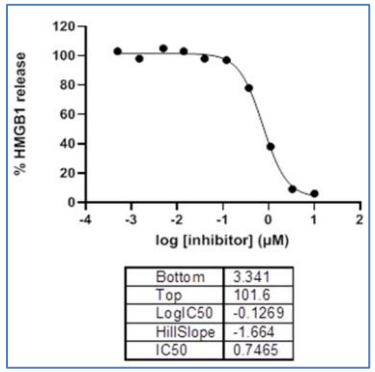
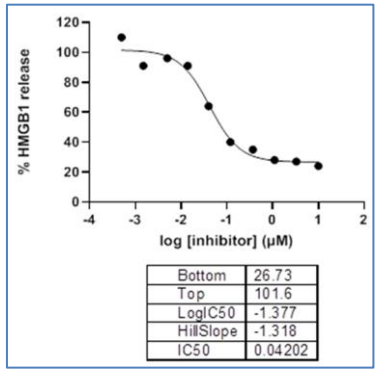
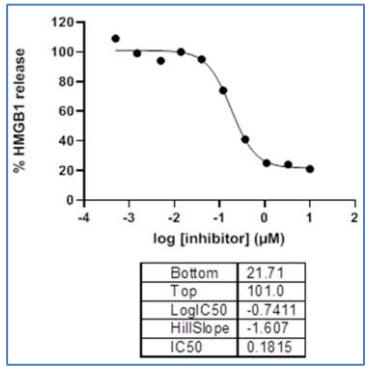
Case Study – 2: *In vitro* Biology Capabilities – TPDs (Functional Assay)



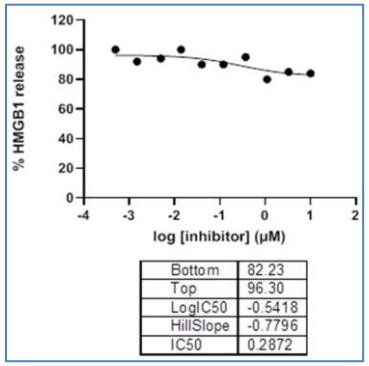
Bottom	3.341
Top	101.6
LogIC50	-0.1269
HillSlope	-1.664
IC50	0.7465



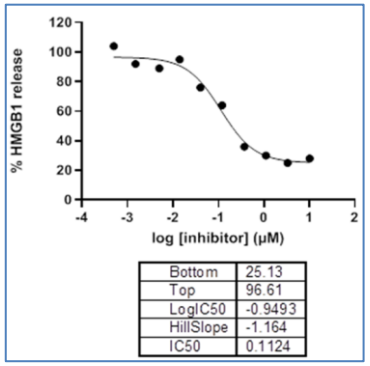
Bottom	26.73
Top	101.6
LogIC50	-1.377
HillSlope	-1.318
IC50	0.04202



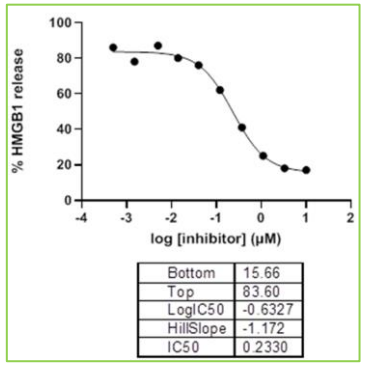
Bottom	21.71
Top	101.0
LogIC50	-0.7411
HillSlope	-1.607
IC50	0.1815



Bottom	82.23
Top	96.30
LogIC50	-0.5418
HillSlope	-0.7796
IC50	0.2872

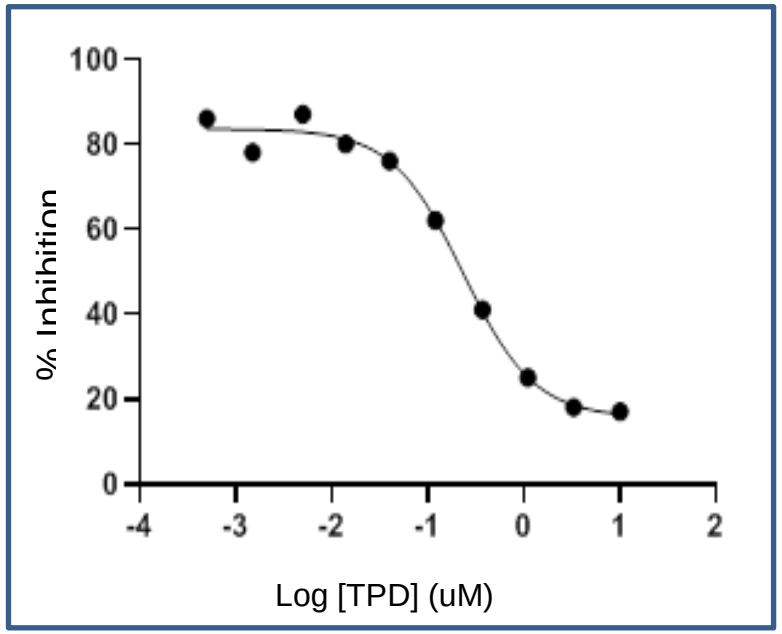


Bottom	25.13
Top	96.61
LogIC50	-0.9493
HillSlope	-1.164
IC50	0.1124

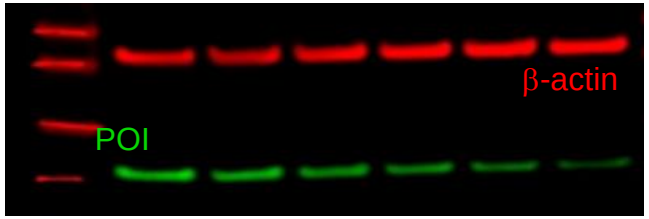


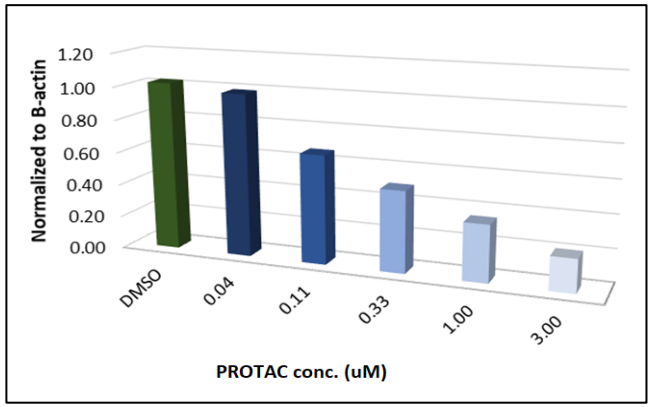
Bottom	15.66
Top	83.60
LogIC50	-0.6327
HillSlope	-1.172
IC50	0.2330

- Selected TPDs were tested in cell-based assays
- TPDs showing <100 nM IC_{50} values was tested in target engagement and degradation assay

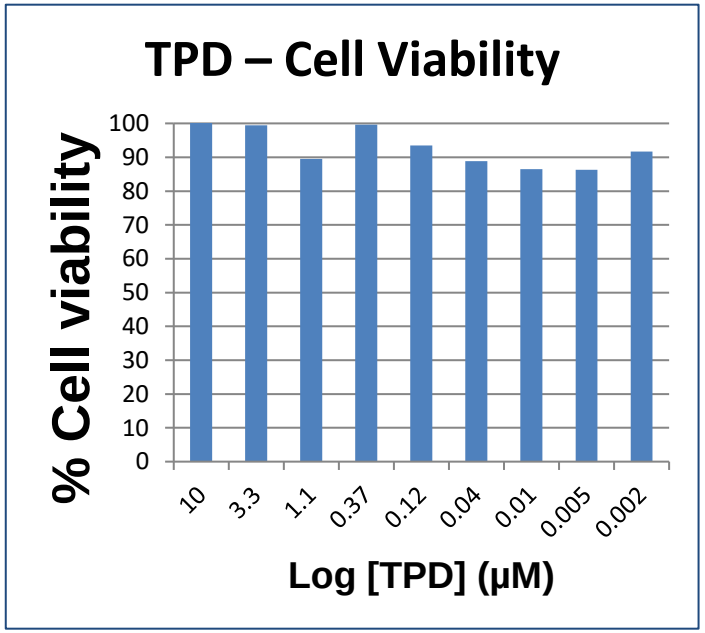


Degradation assay (Western blot)





Cell growth kinetic assay



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